

Exploratory Data Analysis (EDA) Report

Titanic Passenger Dataset | Task 5 - Data Analyst Internship

This report presents a full exploratory data analysis of the Titanic passenger dataset (891 rows, 15 columns). It covers dataset structure, univariate, bivariate and multivariate analysis, correlation and multicollinearity checks, and a final summary of key findings.

1. Dataset Overview

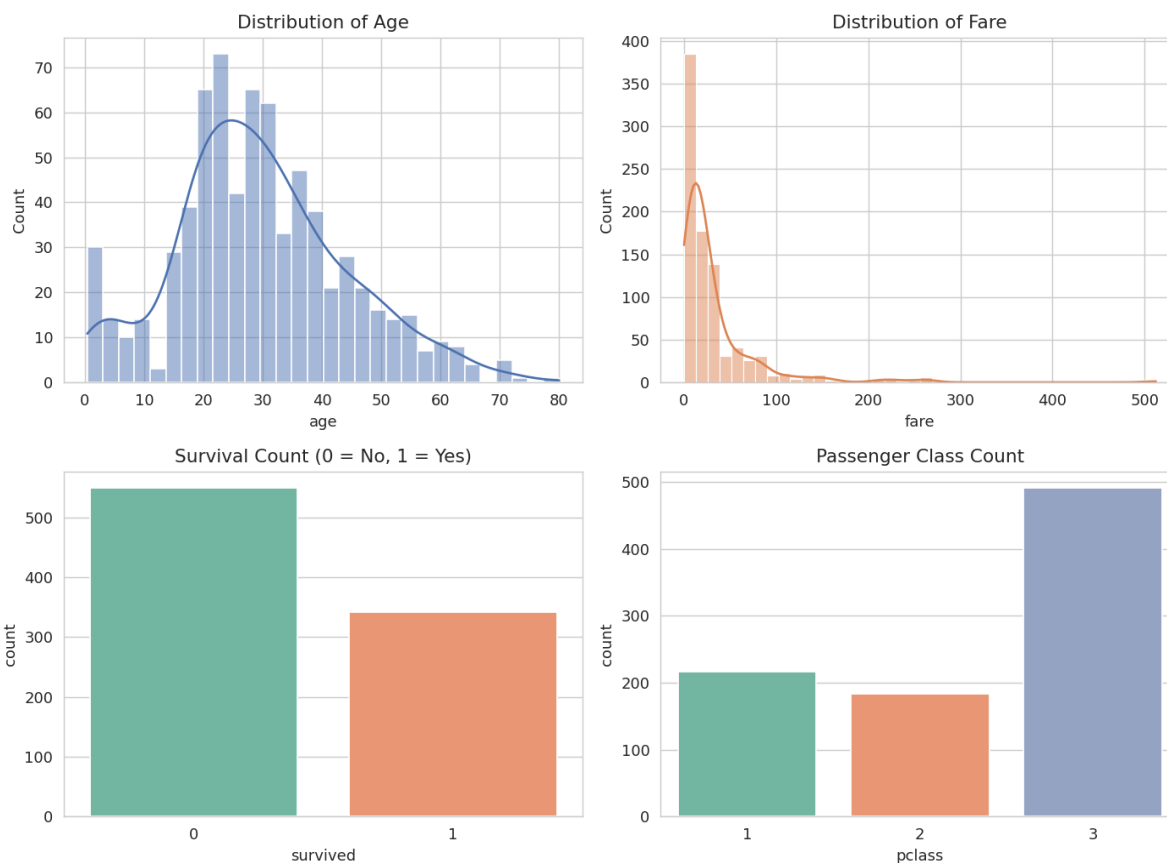
The dataset contains 891 passenger records with 15 columns describing survival status, class, sex, age, fare, family relations, and boarding details.

Metric	Value
Total rows	891
Total columns	15
Missing - Age	177 (19.9%)
Missing - Deck	688 (77.2%)
Missing - Embarked	2 (0.2%)
Survived	342 (38.4%)
Did not survive	549 (61.6%)

Reasoning: Checking shape, data types, and missing values first (using `.info()` and `.describe()`) is essential before any visual analysis - it tells us which columns are usable, which need cleaning, and which should be dropped (e.g. Deck, which is missing for over 77% of passengers).

2. Univariate Analysis - Histograms

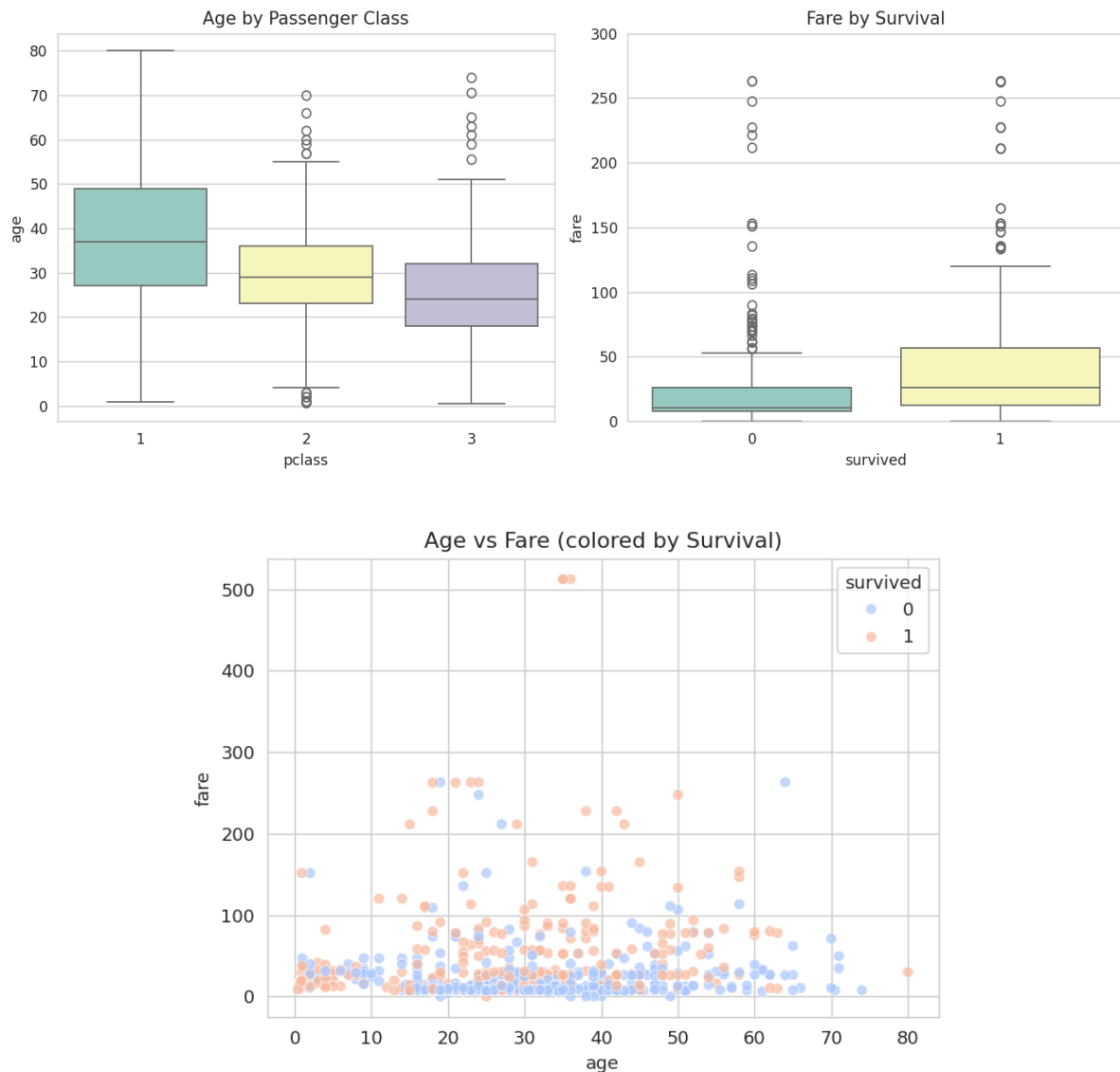
Univariate analysis studies one variable at a time to understand its distribution, spread, and central tendency.



Observations: Age is roughly bell-shaped with a slight right skew (skewness 0.39); most passengers are 20-40 years old. Fare is heavily right-skewed (skewness 4.79) - a few expensive first-class tickets create a long tail. More passengers did not survive (549) than survived (342), and Class 3 had the most passengers (491).

3. Bivariate Analysis - Boxplots & Scatterplots

Bivariate analysis studies the relationship between two variables at a time.

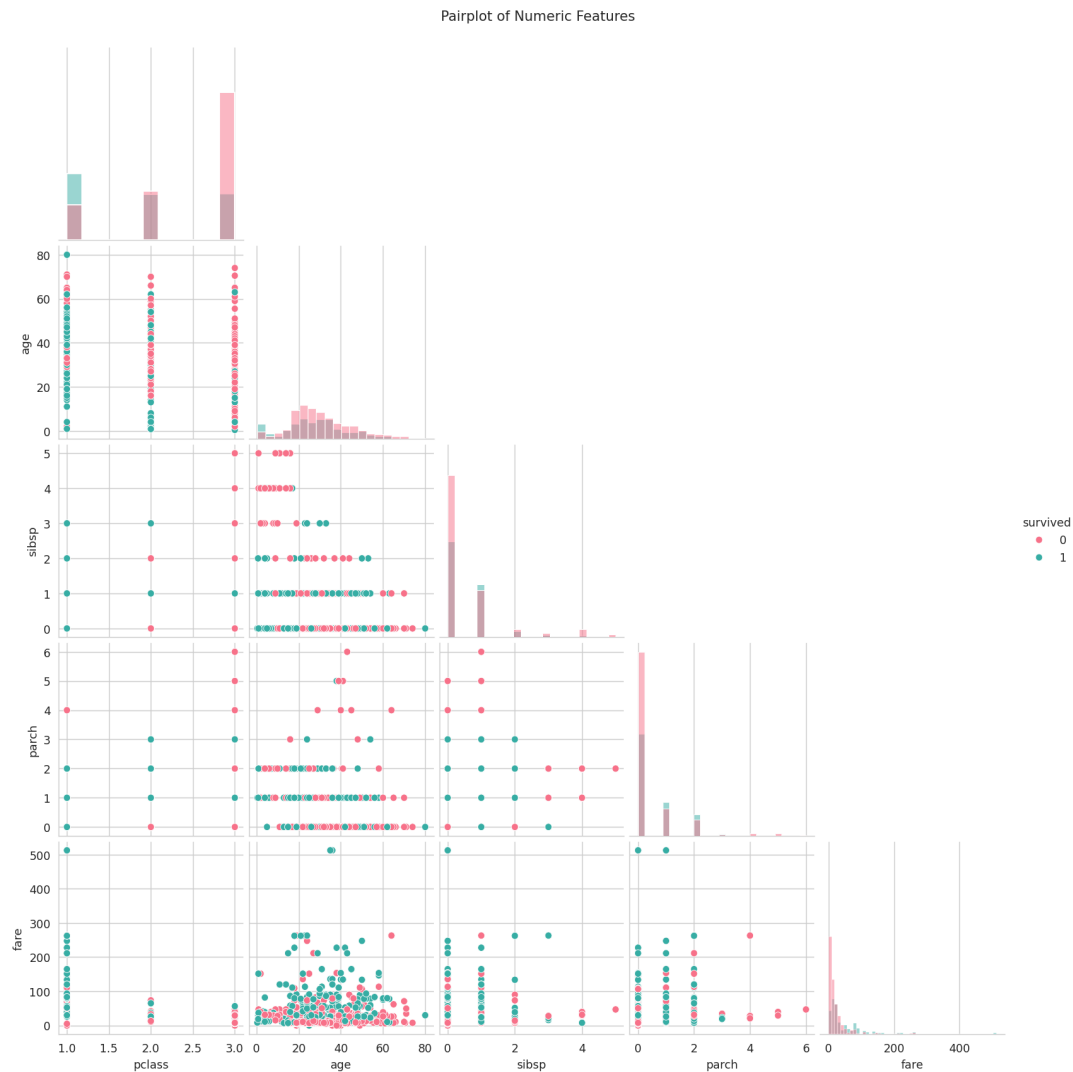


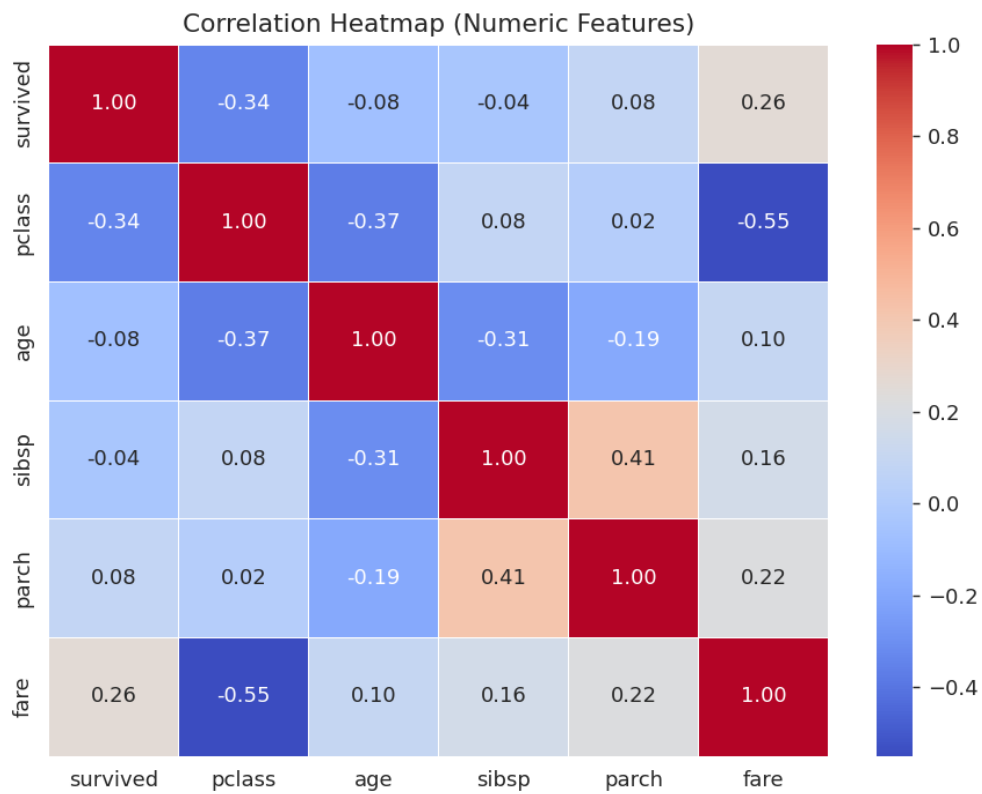
Observations: First-class passengers tend to be older on average than third-class passengers.

Passengers who survived paid somewhat higher fares on average, with several high-fare outliers. Age vs Fare shows no strong linear pattern, but the higher-fare band contains a greater share of survivors.

4. Multivariate Analysis - Pairplot & Correlation Heatmap

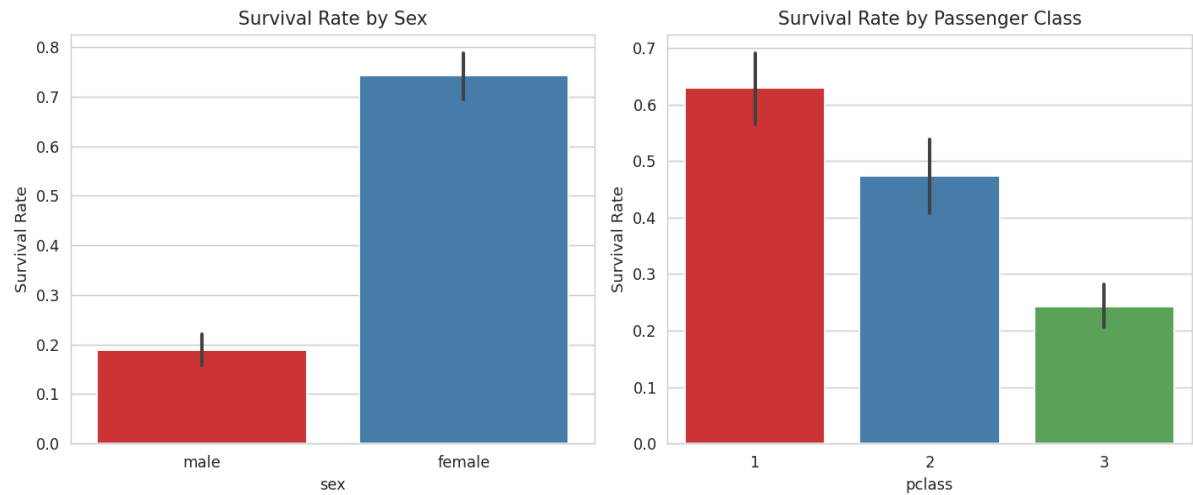
Multivariate analysis studies relationships across three or more variables simultaneously, helping detect broader patterns and multicollinearity.





Observations: Pclass and Fare are strongly negatively correlated (-0.55) - better classes (lower Pclass number) cost more. Survived correlates most with Fare (+0.26) and Pclass (-0.34). SibSp and Parch are moderately correlated (0.41), suggesting families travelled together. No pair exceeds ± 0.8 , so there is **no serious multicollinearity** among the numeric features.

5. Survival Rate Breakdown



Observations: Women survived at 74.2% versus 18.9% for men, reflecting evacuation priority given to women and children. First-class passengers survived at 63.0%, versus 47.3% (2nd class) and 24.2% (3rd class) - likely due to cabin location and closer access to lifeboats.

6. Summary of Findings

1. Only 38.4% of passengers survived overall.
2. Sex was the strongest survival factor - women survived far more often (74.2%) than men (18.9%).
3. Passenger class mattered a great deal - 63.0% survival in 1st class vs 24.2% in 3rd class.
4. Fare and Class are closely linked (correlation -0.55); median fare was Rs.60.29 in 1st class vs Rs.8.05 in 3rd class.
5. Fare is heavily right-skewed, so median (not mean) is the better 'typical value' summary for it.
6. Age has ~20% missing values and Deck has ~77% missing - Age should be imputed (e.g. median) and Deck likely dropped for further modeling.
7. No serious multicollinearity was found among the numeric features (all correlations stayed within ± 0.55), so they can safely be used together in further analysis.